Mrinalini De

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Data Scientist / Data Engineer (M.S. Candidate) using **Python, SQL, and data warehousing** to turn messy data into **ETL pipelines**, **EDA**, and **predictive / clustering models** that support business and product decisions. Experienced working with Product and Engineering and explaining results to non-technical stakeholders; co-author of a **peer-reviewed IJACEN paper** on deep learning–based face recognition.

Skills

Programming & SQL: Python (Pandas, NumPy), SQL (joins, CTEs, window functions)
Machine Learning: Regression, Classification, Tree-based Models (Random Forest, Gradient Boosting), Clustering, Feature Engineering, Cross-validation, Model Evaluation, TensorFlow, Scikit-learn
Analytics: Exploratory Data Analysis (EDA), Product Metrics, Cohort Analysis, Customer Segmentation (RFM)
Data Engineering: ETL Pipelines, Data Modeling, Data Quality Checks, Schema Design, Git
Cloud & Warehousing: AWS (S3, Redshift, QuickSight), Relational Databases (MySQL, PostgreSQL)
Visualization: Power BI, Tableau, Excel

Professional Experience

Data Scientist Intern Aug 2023 – Sep 2023
Zeta Holdings, Saratoga, CA

- Owned end-to-end **ETL pipelines** in **Python (Pandas, NumPy)** and **SQL**, writing parameterized queries, joins, and window functions to clean and integrate messy, real-world customer and transaction data into analysis-ready tables and reusable **SQL views** for analytics.
- Performed **exploratory data analysis (EDA)** on large customer datasets, engineered **RFM-based features**, and applied **unsupervised clustering** and **regression models** to identify high-value customer cohorts, improving campaign targeting by **15%**.
- Built **Power BI dashboards** on top of warehouse tables, defined calculated measures, and partnered with Product Managers and Analysts to surface model-driven insights, reducing reporting turnaround from **2 days to 4 hours** and enabling more data-driven decisions.

Software Development Intern Jun 2022 – Jul 2022
Bhabha Atomic Research Centre, Tarapur, India

- Implemented backend modules in **PHP/MySQL** on a Linux-based stack to capture structured operational data, handling server-side validation, input sanitization, and multi-step form workflows for **500+** internal users.
- Redesigned relational schemas (normalization to 3NF), added appropriate **primary/foreign keys** and indexes, and used **EXPLAIN** to profile slow queries, cutting average page-level response times by **30%**.
- Strengthened **data integrity** by enforcing NOT NULL/UNIQUE constraints, adding defensive checks in application logic, and standardizing error-handling paths to reduce data-entry issues in downstream reporting.

Academic Projects

Deep Learning–Powered Attendance Tracker — MTCNN, FaceNet, DeepFace, TensorFlow Jul 2023 – Apr 2024

- Framed manual attendance tracking as a **face-recognition classification** task; analyzed variations in lighting, pose, and occlusions to guide model and data-augmentation choices.
- Designed a web-based attendance system comparing **MTCNN + FaceNet** and **DeepFace** pipelines using a custom dataset; implemented detection → embedding → verification pipeline and stored attendance in a **relational SQL** warehouse with audit logs.
- Achieved **95% recognition accuracy**, reduced manual effort by **80%**, and published results in *IJACEN*.

Customer Segmentation Using Behavioral Analysis — Python, SQL, Streamlit Jan 2023 – Apr 2023

- Collected and cleaned transaction-level customer data in Python/SQL; performed **EDA** to understand spending distributions and recency/frequency patterns.
- Built a reproducible pipeline to compute **RFM features**, applied **K-means** clustering, and validated clusters using retention and revenue metrics; insights improved targeting by **18%**.
- Developed interactive Streamlit visualizations to communicate cluster profiles and recommended actions to non-technical stakeholders.

Data Pipeline Optimization for Sales Analytics — Python, AWS (S3, Redshift), Tableau May 2024 – Aug 2024

- Consolidated sales data from multiple sources into a centralized **Redshift** warehouse; implemented Python-based ETL with schema validation and **data quality checks**.
- Designed OLAP-friendly tables and optimized SQL aggregations, reducing **dashboard refresh latency by 60%** for recurring product metrics and executive reporting.

Education

Arizona State University Tempe, AZ
*M.S. in Information Technology (**GPA: 3.89 / 4.00**)* *Expected May 2026*
University of Mumbai (FCRIT) Mumbai, India
*B.E. in Information Technology (**GPA: 3.93 / 4.00**)* *May 2024*